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SEMICONDUCTOR STRUCTURE

Abstract

A semiconductor structure includes a dielectric layer, a first through via, a second through via, and a third through via. The dielectric layer is disposed in an interconnect structure, and includes a first doping region and a second doping region, wherein a first doping concentration of the first doping region is different from a second doping concentration of the second doping region. The first through via is surrounded by a dopant-free region of the dielectric layer. The second through via is surrounded by the first doping region. The third through via is surrounded by the second doping region.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This patent application is a divisional application of U.S. patent application Ser. No. 17/724,210, filed on Apr. 19, 2022, entitled of “METHOD FOR MANUFACTURING THROUGH VIAS USING AMORPHIZATION TO ADJUST ETCHING RATE”, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] Semiconductor devices are used in a variety of electronic applications, such as personal computers, cellular phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductive layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon. As the semiconductor industry has progressed into advanced technology process nodes in pursuit of greater device density, higher performance, and lower costs, challenges of precise control of etching rates across a wafer have arisen.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a flow diagram of a method for manufacturing a semiconductor structure in accordance with some embodiments of the disclosure.

[0005] FIG. 2 is a flow diagram of a method for manufacturing a semiconductor structure in accordance with some embodiments of the disclosure.

[0006] FIG. 3 is a schematic top view of a semiconductor structure in accordance with some embodiments of the disclosure.

[0007] FIGS. 4 to 8 are schematic diagrams at different stages of a method for manufacturing a semiconductor structure in accordance with different embodiments of the disclosure.

[0008] FIG. 9 is a schematic top view of a mask layer in accordance with some embodiments of the disclosure.

[0009] FIGS. 10 to 11 are schematic diagrams at different stages of a method for manufacturing a semiconductor structure in accordance with different embodiments of the disclosure.

[0010] FIG. 12 is a schematic top view of a mask layer in accordance with some embodiments of the disclosure.

[0011] FIGS. 13 to 18 are schematic diagrams at different stages of a method for manufacturing a semiconductor structure in accordance with different embodiments of the disclosure.

[0012] FIG. 19 is a schematic diagram at a stage of a method for manufacturing a semiconductor structure in accordance with different embodiments of the disclosure.

[0013] FIG. 20 is a schematic top view of a mask layer in accordance with some embodiments of

the disclosure.

[0014] FIGS. **21** to **24** are schematic diagrams at different stages of a method for manufacturing a semiconductor structure in accordance with different embodiments of the disclosure.

[0015] FIGS. **25** to **26** are schematic diagrams at different stages of a method for manufacturing a semiconductor structure in accordance with different embodiments of the disclosure.

[0016] FIG. **27** is a schematic top view of a mask layer in accordance with some embodiments of the disclosure.

[0017] FIG. **28** is a schematic diagram at a stage of a method for manufacturing a semiconductor structure in accordance with different embodiments of the disclosure.

DETAILED DESCRIPTION

[0018] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of elements and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0019] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “upper,” “on” and the like, may be used herein for case of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0020] As used herein, although the terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context.

[0021] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately” and “about” generally mean within a value or range that can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately” and “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the

number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

[0022] FIG. 1 is a flow diagram of a method **600** for manufacturing a semiconductor structure in accordance with some embodiments of the present disclosure. The method **600** includes a number of operations (**601**, **602**, **603** and **604**) and the description and illustration are not deemed as a limitation to the sequence of the operations. A first layer is formed over a substrate in a first region and a second region in the operation **601**. A first etching with a first etching rate is performed on the first layer in the operation **602**, thereby forming a first trench in the first region and a second trench in the second region. An amorphization **51** is performed on the first layer in the second region to form a third layer in the operation **603**, wherein the amorphization **51** targets a bottom of the second trench. A second etching is performed on the third layer with a second etching rate in the operation **604**, wherein the second etching rate is greater than the first etching rate. It should be noted that the operations of the method **600** may be rearranged or otherwise modified within the scope of the various aspects. Additional processes may be provided before, during, and after the method **600**, and some other processes may be only briefly described herein. Thus, other implementations are possible within the scope of the various aspects described herein.

[0023] FIG. 2 is a flow diagram of a method **700** for manufacturing a semiconductor structure in accordance with some embodiments of the present disclosure. The method **700** includes a number of operations (**701**, **702**, **703**, **704**, **705**, **706**, **707** and **708**) and the description and illustration are not deemed as a limitation to the sequence of the operations. A dielectric layer is formed over a substrate in the operation **701**. Portions of the dielectric layer are removed in the operation **702**, thereby forming a first opening in a first region, a second opening in a second region and a third opening in a third region. A first mask layer is formed in the first region in the operation **703** to cover the first opening. A first dopant is introduced to a bottom of the second opening and a bottom of the third opening in the operation **704**. A second mask layer is formed in the second region in the operation **705** to cover the second opening. A second dopant is introduced to the bottom of the third opening in the operation **706**. The first mask layer and the second mask layer are removed in the operation **707**. The dielectric layer is etched in the operation **708**, thereby forming a first through via, a second through via, and a third through via penetrating the dielectric layer. It should be noted that the operations of the method **700** may be rearranged or otherwise modified within the scope of the various aspects. Additional processes may be provided before, during, and after the method **700**, and some other processes may be only briefly described herein. Thus, other implementations are possible within the scope of the various aspects described herein.

[0024] The method **600** and the method **700** are within the same concept of the present disclosure, and in order to further illustrate details of the method **600**, the method **700**, and the concept of the present disclosure, the method **600** and the method **700** are comprehensively illustrated with embodiments of the present disclosure.

[0025] Please refer to FIG. 3, which is a schematic diagram of a semiconductor structure **10** from a top view perspective in accordance with some embodiments of the present disclosure. The semiconductor structure **10** can be a die, a chip, or an active area of a wafer. The semiconductor structure **10** may include multiple regions. In some embodiments as shown in FIG. 3, the semiconductor structure **10** includes a first region **11**, a second region **12**, and a third region **13**. In some embodiments, the first region **11**, the second region **12** and the third region **13** have different densities of active components. In some embodiments, the active components include transistors. In some embodiments, the first region **11**, the second region **12** and the third region **13** have different densities of electric paths. In some embodiments, a greater density of electric paths includes a greater density of conductive components of an interconnect structure. In some embodiments, a distance between adjacent conductive components of the interconnect structure decreases as a density of electric paths increases.

[0026] Please refer to FIG. 4, which is a schematic cross-sectional diagram along a line A-A' of the semiconductor structure **10** shown in FIG. 3 at a stage of the method **600** or the method **700** in accordance with some embodiments of the present disclosure. Prior to the operation **601** or the operation **701**, a substrate **14** is provided, received, or formed. In some embodiments, the substrate **14** includes a plurality of electrical components **142** formed on a semiconductive layer **141**. The semiconductive layer **141** (which may be referred to as a die substrate) includes a bulk semiconductor material, such as silicon. In one embodiment, the substrate **14** may include other semiconductor materials, such as silicon germanium, silicon carbide, gallium arsenide, or the like. The semiconductive layer **141** may be of a first conductivity type, e.g., a P-type semiconductive substrate (acceptor type), or of a second conductivity type, e.g., an N-type semiconductive substrate (donor type). Alternatively, the semiconductive layer **141** may include another elementary semiconductor, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, or GaInAsP; or combinations thereof. In another embodiment, the semiconductive layer **141** may be a semiconductor-on-insulator (SOI). In still other embodiments, the semiconductive layer **141** may include a doped epitaxial layer, a gradient semiconductor layer, or a semiconductor layer overlaying another semiconductor layer of a different type, such as a silicon layer on a silicon germanium layer.

[0027] The plurality of electrical components **142** are formed on the semiconductive layer **141** following conventional methods of manufacturing semiconductors. The electrical components **142** can be active components or devices, and may include different types or generations of devices. The electrical components **142** can include a planar transistor, a multi-gate transistor, a gate-all-around field-effect transistor (GAAFET), a fin field-effect transistor (FinFET), a vertical transistor, a nanosheet transistor, a nanowire transistor, or a combination thereof. For a purpose of simplicity, a planar transistor is depicted in FIG. 4 as an exemplary embodiment of an electrical component, but such depiction is not intended to limit the present disclosure.

[0028] The substrate **14** may further include an insulating layer **143** formed over the semiconductive layer **141**, and a plurality of contacts **144** formed in the insulating layer **143** and electrically connected to the electrical components **142**. In some embodiments, the contacts **144** are connected to a source region, a drain region and a gate region of a transistor (which can be one of the electrical components **142**). In some embodiments, the first region **11**, the second region **12**, and the third region **13** are defined by different densities of electrical components **142**. In some embodiments, a first density of the electrical components **142** in the first region **11** is greater than a second density of the electrical components **142** in the second region **12**. In some embodiments, the second density of the electrical components **142** in the second region **12** is greater than a third density of the electrical components **142** in the third region **13**.

[0029] The substrate **14** may further include a plurality of conductive elements **146**. In some embodiments, the conductive elements **146** are electrically isolated by an insulating layer **145**. In some embodiments, a material of the conductive elements **146** includes copper (Cu), aluminum (Al), tungsten (W), manganese (Mn), cobalt (Co), titanium (Ti), or a combination thereof. In some embodiments, the insulating layer **145** includes silicon oxide, silicon nitride, silicon oxy-nitride, a low-k dielectric material, a high-k dielectric material, combinations thereof, and/or other suitable materials. The conductive elements **146** may be a layer of silicides, pads, traces, vias, plugs, plates, or contacts, or a layer having a combination thereof. In some embodiments as shown in FIG. 4, the conductive elements **146** are electrically connected to the contacts **144**. In some embodiments, the conductive elements **146** collectively comprise a first conductive layer formed above the contacts **144**. The conductive elements **146** can be comprehensively referred to as a first conductive layer of an interconnect structure. In some embodiments, the conductive elements **146** are disposed in another layer above the first conductive layer of the interconnect structure.

[0030] Densities of the conductive elements **146** in different regions may correspond to the densities of the electrical components **142** in the different regions. A density of the conductive elements **146** can be defined by an area ratio of a total area of the conductive elements **146** to a total area of the insulating layer **145** in one region (e.g., the first region **11**, the second region **12** or the third region **13**) from a top view perspective. In some embodiments, the total area of the conductive elements **146** is defined by an exposed surface area of the conductive elements **146**. In some embodiments, the total area of the insulating layer **145** is defined by an exposed surface area of the insulating layer **145**. In some embodiments, a first density of the conductive elements **146** in the first region **11** is greater than a second density of the conductive elements **146** in the second region **12**. In some embodiments, the second density of the conductive elements **146** in the second region **12** is greater than a third density of the conductive elements **146** in the third region **13**. Widths of the conductive elements **146** in different regions may correspond to the densities of the electrical components **142** in the different regions. In some embodiments, a first width W1 of each of the conductive elements **146** in the first region **11** is less than a second width W2 of each of the conductive elements **146** in the second region **12**. In some embodiments, the second width W2 is less than a third width W3 of each of the conductive elements **146** in the third region **13**. A distance between two adjacent electrical elements **146** in one region may correspond to a density of the electrical components **142** of the region. A lower density of the electrical components **142** in one region can correspond to a greater distance between two adjacent conductive elements **146** in such region. In some embodiments, a first distance D1 between two adjacent conductive elements **146** in the first region is less than a second distance D2 between two adjacent conductive elements **146** in the second region. In some embodiments, the second distance D2 is less than a third distance D3 between two adjacent conductive elements **146** in the third region. In some embodiments, the width W1, the width W2 or the width W3 is in a range of 0.0001 to 500 microns.

[0031] For case of illustration and simplicity of drawings, portions of the substrate **14** below the insulating layer **145** and the conductive elements **146** are omitted in the following disclosure. However, it is not intended to limit the present disclosure.

[0032] Please refer to FIG. 5, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 in accordance with the operation **601** of the method **600** and/or the operation **701** of the method **700** according to some embodiments of the present disclosure. A dielectric layer **152** is formed over the substrate **14**. More specifically, the dielectric layer **152** may be formed on and may cover the conductive elements **146** and the insulating layer **145**. In some embodiments, the dielectric layer **152** is formed by a deposition. In some embodiments, the dielectric layer **152** is formed by a blanket deposition. In some embodiments, the dielectric layer **152** covers an entirety of the conductive elements **146** in the first region **11**, the second region **12** and the third region **13**. In some embodiments, an etch stop layer **151** is formed prior to the formation of the dielectric layer **152** across the substrate **14**. The etch stop layer **151** may be formed over the substrate **14** by a deposition. In some embodiments, the etch stop layer **151** covers at least the conductive elements **146** in the first region **11**, the second region **12** and the third region **13**. In some embodiments, the etch stop layer **151** is formed by a blanket deposition.

[0033] Please refer to FIG. 6, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. Prior to the operation **602** of the method **600** and/or the operation **702** of the method **700**, the method **600** and/or the method **700** may further include operations of forming and patterning a mask layer **16**. In some embodiments, the mask layer **16** is a multi-layered structure as shown in FIG. 6. In some embodiments, the mask layer **16** includes a first sublayer **161**, a second sublayer **162** and a third sublayer **163**. The first sublayer **161**, the second sublayer **162** and the third sublayer **163** may have different patterns. In some embodiments as shown in FIG. 6, a plurality of openings **17** defined by the first sublayer **161** are overlapped by a plurality of openings **18** defined by the second sublayer **162** and the third sublayer **163**. In some

embodiments, each of the openings **17** connects to and extends from each of the openings **18**. In some embodiments, the opening **17** is smaller than its corresponding opening **18**. In some embodiments, the opening **17** is entirely overlapped by the corresponding opening **18** to form a T-shaped trench. A plurality of the T-shaped trenches may be used to define openings and through vias formed in the dielectric layer **152** in subsequent operations of the method **600** and/or the method **700**.

[0034] Widths of the openings **17** in different regions can be different. In some embodiments, the openings **17** include openings **171** in the first region **11**, openings **172** in the second region **12**, and openings **173** in the third region **13**. In some embodiments, a width **W4** of an opening **171** in the first region **11** is less than a width **W5** of an opening **172** in the second region **12**. In some embodiments, the width **W5** of the opening **172** is less than a width **W6** of an opening **173** in the third region **13**. Widths of the openings **18** in different regions can be also different. In some embodiments, the openings **18** include openings **181** in the first region **11**, openings **182** in the second region **12**, and openings **183** in the third region **13**. In some embodiments, a width **W7** of an opening **181** in the first region **11** is less than a width **W8** of an opening **182** in the second region **12**. In some embodiments, the width **W8** of the opening **182** is less than a width **W9** of an opening **183** in the third region **13**.

[0035] Distances between the openings **18** within a region can vary between regions. In some embodiments, a distance **D4** between adjacent openings **181** in the first region **11** is less than a distance **D5** between adjacent openings **182** in the second region **12**. In some embodiments, the distance **D5** is less than a distance **D6** between adjacent openings **183** in the third region **13**. Distances between the openings **17** within a region can also vary between regions, and can correspond to the distances between the openings **18** in different regions as described above. Densities of the openings **18** (or the openings **17**) in different regions correspond to densities of the conductive elements **146** and/or densities of the electrical elements **142** as illustrated above. Repeated description is omitted herein.

[0036] Please refer to FIG. 7, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. In the operation **602** of the method **600** and/or the operation **702** of the method **700**, a first etching is performed, and portions of the dielectric layer **152** are removed using the mask layer **16** as a mask. In some embodiments, the first etching targets the dielectric layer **152**, thereby forming a plurality of openings **19** in different regions (e.g., the first region **11**, the second region **12** and the third region **13**) in the dielectric layer **152**. The plurality of openings **19** can include a plurality of first openings **191** in the first region **11**, a plurality of second openings **192** in the second region **12**, and a plurality of third openings **193** in the third region **13**. In some embodiments, the plurality of first openings **191** may be defined by the openings **171** of the mask layer **16**. In some embodiments, the plurality of second openings **192** may be defined by the openings **172** of the mask layer **16**. In some embodiments, the plurality of third openings **193** may be defined by the openings **173** of the mask layer **16**. In some embodiments, the first openings **191**, the second openings **192** and the third openings **193** are respectively referred to as first trenches **191**, second trenches **192** and third trenches **193**.

[0037] At least one of the first openings **191**, the second openings **192** and the third openings **193** may stop at the dielectric layer **152** and may be separated from the conductive elements **146**. Etching rates of the first etching in different regions may be different. In some embodiments, the etching rate of the first etching in each region corresponds to the densities of the openings **17** or the openings **18** in such region. In some embodiments, the etching rate of the first etching in each region corresponds to the widths of the openings **17** or the openings **18** in such region. In some embodiments, the etching rate of the first etching in each region corresponds to the distances between the openings **18** in such regions. In some embodiments, an etching rate in the first region is greater than an etching rate in the second region. In some embodiments, the etching rate in the

second region is greater than an etching rate in the third region. In other words, in such embodiments, a depth H1 of the first openings 191 measured from a top of the dielectric layer 152 to a bottom of the first openings 191 is greater than a depth H2 of the second openings 192 measured from the top of the dielectric layer 152 to a bottom of the second openings 192. In some embodiments, the depth H2 is greater than a depth H3 of the third openings 193 measured from the top of the dielectric layer 152 to a bottom of the third openings 193.

[0038] Please refer to FIG. 8, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method 600 and/or the method 700 according to some embodiments of the present disclosure. Prior to the operation 603 of the method 600 and/or in the operation 703 of the method 700, a photoresist layer 21 is formed in the first region 11 over the substrate 14. In some embodiments, the photoresist layer 21 is referred to as a first mask layer 21. In some embodiments, the first mask layer 21 can include a positive photoresist material, a negative photoresist material, oxide, nitride, oxynitride, another suitable hard mask material, or a combination thereof. In some embodiments, a material of the first mask layer 21 is different from those of the mask layer 16 and that of the dielectric layer 152. For case of description, the term "photoresist layer" is used to describe the layer 21 shown in the figures. The photoresist layer 21 may cover all of the first openings 191 in the first region 11 and leave exposed the second openings 192 in the second region and the third openings 193 in the third region 13. The photoresist layer 21 may or may not fill all portions of the first openings 191. In some embodiments, the photoresist layer 21 fills the first opening 191. In some embodiments, one or more of the first openings 191 are partially filled by the photoresist layer 21. In some embodiments, the first openings 191 are completely filled by the photoresist layer 21. In some embodiments, the photoresist layer 21 covers or seals tops of the first openings 191.

[0039] FIG. 9 is a schematic top view of the photoresist layer 21 at the stage shown in FIG. 8 of the method 600 and/or the method 700 according to some embodiments of the present disclosure. As shown in FIG. 9 and FIG. 3, the photoresist layer 21 may be formed in only the first region 11 of the semiconductor structure 10. In some embodiments, a configuration of the photoresist layer 21 is substantially same as a configuration of the first region 11 from a top view perspective. In some embodiments, the photoresist layer 21 leaves a portion of the first region 11 exposed and covers all the first openings 191. In some embodiments, the photoresist layer 21 covers a portion of the substrate 14 outside the first region 11 but leaves exposed an entirety of the second openings 192 and an entirety of the third openings 193.

[0040] Please refer to FIG. 10, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method 600 and/or the method 700 according to some embodiments of the present disclosure. In the operation 603 of the method 600 and/or the operation 704 of the method 700, an amorphization 51 is performed on the dielectric layer 152. In some embodiments, the amorphization 51 includes an introduction of a first dopant to the dielectric layer 152. In some embodiments, the amorphization 51 is realized using a first implantation. In some embodiments, the amorphization 51 is performed in the second region 12 and the third region 13. In some embodiments, the amorphization 51 targets bottoms of the second openings 192 and the third openings 193. In some embodiments, the first dopant is introduced to the bottom of each of the second openings 192 and the bottom of each of the third openings 193. In some embodiments, the second dopant includes carbon (C), silicon (Si), germanium (Ge), tin (Sn), lead (Pb), boron (B), aluminum (Al), gallium (Ga), indium (In), thallium (Tl), nitride (N), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi), or a combination thereof. A first region 152a of the dielectric layer 152 surrounding bottom portions of each of the first openings 191 is denser than a second region 152b of the dielectric layer 152 surrounding bottom portions of each of the second openings 192. A third region 152c of the dielectric layer 152 surrounding each of bottom portions of the third openings 193 is less dense than the first region 152a of the dielectric layer 152. A structural density of the third region 152c is substantially equal to a structural density of the second region 152b. In

some embodiments, a doping concentration of the second region **152b** is substantially equal to a doping concentration of the third region **152c**. In some embodiments, a boundary may be formed or observed between the second region **152b** and a remaining portion of the dielectric layer **152**. In some embodiments, a boundary may be formed or observed between the third region **152c** and the remaining portion of the dielectric layer **152**. In some embodiments, the first region **152a** of the dielectric layer **152** is dopant-free.

[0041] Please refer to FIG. **11**, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. **3** at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. Prior to the operation **604** of the method **600** and/or in the operation **705** of the method **700**, a photoresist layer **22** is formed in the second region **12** over the substrate **14**. In some embodiments, the photoresist layer **22** is referred to as a second mask layer **22**. A material of the second mask layer **22** can be similar to or same as that of the first mask layer **21** as illustrated above. For case of description, the term "photoresist layer" is used to describe the layer **22** shown in the figures. The photoresist layer **22** may cover all of the second openings **192** in the second region **12** and leave exposed the third openings **193** in the third region **13**. The photoresist layer **22** may or may not fill all portions of the second openings **192**. In some embodiments, the photoresist layer **22** fills the second opening **192**. In some embodiments, one or more of the second openings **192** are partially filled by the photoresist layer **22**. In some embodiments, the second openings **192** are completely filled by the photoresist layer **22**. In some embodiments, the photoresist layer **22** covers or seals tops of the first openings **191**.

[0042] FIG. **12** is a schematic top view of the photoresist layer **22** at the stage shown in FIG. **11** of the method **600** and/or the method **700** according to some embodiments of the present disclosure. Referring to FIG. **12** and FIG. **3**, the photoresist layer **22** may be formed in only the second region **12** of the semiconductor structure **10**. In some embodiments, a configuration of the photoresist layer **22** is substantially same as a configuration of the second region **12** from a top view perspective. In some embodiments, the photoresist layer **22** leaves a portion of the second region **12** exposed and covers all the second openings **192**. In some embodiments, the photoresist layer **22** covers a portion of the substrate **14** outside the second region **12** but leaves an entirety of the third openings **193** exposed.

[0043] Please refer to FIG. **13**, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. **3** at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. Prior to the operation **604** of the method **600** and/or in the operation **706** of the method **700**, an amorphization **52** is performed on the dielectric layer **152**. In some embodiments, the amorphization **52** includes an introduction of a second dopant to the dielectric layer **152**. In some embodiments, the amorphization **52** is realized by using a second implantation. In some embodiments, the amorphization **52** is performed in the third region **13**. In some embodiments, the amorphization **52** targets bottoms of the third openings **193**. In some embodiments, the second dopant is introduced to the third region **152c** of the dielectric layer **152**. The second dopant can be same as or different from the first dopant. In some embodiments, the second dopant includes carbon (C), silicon (Si), germanium (Ge), tin (Sn), lead (Pb), boron (B), aluminum (Al), gallium (Ga), indium (In), thallium (Tl), nitride (N), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi), or a combination thereof. In some embodiments, a structural density of the third region **152c** of the dielectric layer **152** is less than the structural densities of the second region **152b** or the first region **152a** of the dielectric layer **152**. In some embodiments, a doping concentration of the third region **152c** of the dielectric layer **152** is substantially greater than the doping concentration of the second region **152b** of the dielectric layer **152**. In some embodiments, the first region **152a** remains dopant-free. In some embodiments, the second dopant stops in the third region **152c** due to the boundary formed in the operation **603** of the method **600** and/or the operation **704** of the method **700** as shown in FIG. **10**.

[0044] Please refer to FIG. **14**, which is a schematic cross-sectional diagram along the line A-A'

shown in FIG. 3 at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. Prior to the operation **604** of the method **600** and/or in the operation **707** of the method **700**, the photoresist layer **21** and the photoresist layer **22** are removed. The first openings **191**, the second openings **192** and the third openings **193** are thereby exposed. [0045] Please refer to FIG. 15, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. In the operation **604** of the method **600** and/or the operation **708** of the method **700**, a second etching is performed on the dielectric layer **152**. In some embodiments, the third sublayer **163** is used as a mask to remove portions of the dielectric layer **152**. Due to the structural densities of the second region **152b** and the third region **152c** are less than a structural density of a portion of the dielectric layer **152** outside the regions **152b** and **152c**, an etching rate of the second etching **52** in a region **11**, **12** or **13** is greater than an etching rate of the first etching **51** in the same region **11**, **12** or **13**. In addition, a lower structural density can result in a higher etching rate of a same etching operation. In some embodiments, an etching rate of the second etching **52** in the second region **12** is greater than an etching rate of the first etching **51** in the second region **12**. In some embodiments, an etching rate of the second etching **52** in the third region **13** is greater than an etching rate of the first etching **51** in the third region **13**. In some embodiments, an etching rate of the second etching **52** in the third region **13** is greater than an etching rate of the second etching **52** in the second region **12**. In some embodiments, the etching rate of the second etching **52** in the second region **12** is greater than an etching rate of the second etching **52** in the first region **11**. In some embodiments, portions of the first sublayer **161** exposed through the third sublayer **163** are also removed by the second etching. A plurality of through vias **31** are formed in the first region **11**, the second region **12** and the third region **13** through the plurality of openings **19** shown in FIG. 14. In some embodiments, a profile of the through via **31** is similar to a combined profile of a corresponding opening **19**, a corresponding opening **17** and a corresponding opening **18**. In some embodiments, the profile of the through via **31** is defined by the combined profile of the corresponding opening **19**, the corresponding opening **17** and the corresponding opening **18**. In some embodiments, each of the through vias **31** has a T-shaped profile.

[0046] Each of the through vias **31** penetrates the dielectric layer **152**. In some embodiments, each of the through vias **31** penetrates the etch stop layer **151**. In some embodiments, the conductive elements **146** are exposed through the dielectric layer **152** through the through vias **31**. In some embodiments, a portion of the conductive element **146** at a bottom of each of the through vias **31** is removed. The through vias **31** can include a plurality of first through vias **311** in the first region **11**, a plurality of second through vias **312** in the second region **12**, and a plurality of third through vias **313** in the third region **13**. In some embodiments, each of the first through vias **311** penetrates a corresponding first region **152a** of the dielectric layer **152**. In some embodiments, each of the second through vias **312** penetrates a corresponding second region **152b** of the dielectric layer **152**. In some embodiments, each of the third through vias **313** penetrates a corresponding third region **152c** of the dielectric layer **152**. The first region **152a** may surround or encircle a lower portion of each of the first through vias **311**. The second region **152b** may surround or encircle a lower portion of each of the second through vias **312**. The third region **152c** may surround or encircle a lower portion of each of the third through vias **313**. In some embodiments, the lower portion of the through via **311**, **312** or **313** is defined as a vertical portion below a horizontal portion of the T-shaped configuration.

[0047] Please refer to FIG. 16, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. After the operation **604** of the method **600** and/or the operation **708** of the method **700**, the mask layer **16** is removed. A top surface of the dielectric layer **152** is thereby exposed.

[0048] Please refer to FIG. 17, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method 600 and/or the method 700 according to some embodiments of the present disclosure. After the operation 604 of the method 600 and/or the operation 708 of the method 700, a conductive material 41 is formed over the dielectric layer 152. In some embodiments, the conductive material 41 may be formed on the dielectric layer 152 and fills the through vias 31. In some embodiments, the conductive material 41 is electrically connected to the conductive elements 146. In some embodiments, the conductive material 41 is in contact with the conductive elements 146. In some embodiments, the conductive material 41 is in physical contact with the conductive elements 146. A material of the conductive material 41 can be same as or different from that of the conductive elements 146. In some embodiments, the conductive material 41 includes copper (Cu), aluminum (Al), tungsten (W), manganese (Mn), cobalt (Co), titanium (Ti), or a combination thereof.

[0049] Please refer to FIG. 18, which is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method 600 and/or the method 700 according to some embodiments of the present disclosure. After the operation 604 of the method 600 and/or the operation 708 of the method 700, a portion of the conductive material 41 disposed on top of the dielectric layer 152 is removed, and a plurality of via structures are formed in the dielectric layer 152. In some embodiments, a plurality of via structures 411 are formed in the first region 11, wherein each of the via structures 411 is surrounded by a dopant-free region (i.e., the first region 152a). In some embodiments, a plurality of via structures 412 are formed in the second region 12, wherein each of the via structures 412 is surrounded by a first doping region (i.e., the second region 152b). In some embodiments, a plurality of via structures 413 are formed in the third region 13, wherein each of the via structures 413 is surrounded by a second doping region (i.e., the third region 152c). In some embodiments, a doping concentration of the first doping region 152b of the dielectric layer 152 is substantially greater than a doping concentration of the second doping region 152c of the dielectric layer 152. In some embodiments, the second dopant is same as the first dopant, and the second doping region 152c includes a same ingredient as the first doping region 152b. In some embodiments, the second dopant is different from the first dopant, the second doping region 152c includes an ingredient different from that of the first doping region 152b, and a concentration of the first dopant in the first doping region 152b is substantially equal to a concentration of the second dopant in the second doping region 152c.

[0050] As a property of an etching operation, different etching rates may be found in different regions in which via structures of different densities are to be formed. The present disclosure therefore provides a method to control structural densities in different regions according to different densities of the via structures to be formed. A region determined to have a slower etching rate is provided with amorphization one or more times. Therefore, the etching rate in the region can be adjusted so as to allow the etching rates in different regions across a wafer to be substantially equal. A result of the etching operation can be thereby improved, and a product yield can be also improved.

[0051] The above description is for a purpose of illustration of the concept of the present disclosure, and the present disclosure is not limited to the embodiments as described above and illustrated in FIGS. 4 to 18. In order to achieve the purpose of the present disclosure as described above, in an alternative embodiment, the operation 603 of the method 600 depicted in FIG. 10 can be performed after the operation depicted in FIG. 13. Similarly, in an alternative embodiment, the operation 704 of the method 700 depicted in FIG. 10 can be performed after the operation 706 depicted in FIG. 13.

[0052] FIG. 19 is a schematic cross-sectional diagram along the line A-A' shown in FIG. 3 at a different stage of the method 600 and/or the method 700 according to some embodiments of the present disclosure. FIG. 20 is a schematic top view of a photoresist layer 23 according to some embodiments of the present disclosure. Referring to FIGS. 3 and 19 to 20, in alternative

embodiments, after the operation **602** of the method **600** and/or the operation **702** of the method **700**, the photoresist layer **23** is formed over the dielectric layer **152**. In some embodiments, the photoresist layer **23** covers the first region **11** and the second region **12**. In some embodiments, the photoresist layer **23** fills the first openings **191** in the first region **11** and the second openings **192** in the second region **12**. In some embodiments, only the third region **13** is exposed through the photoresist layer **23**.

[0053] FIG. **21** is a schematic cross-sectional diagram along the line A-A' shown in FIG. **3** at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. The operation of the method **600** as depicted in FIG. **13** and/or the operation **706** is performed after formation of the photoresist layer **23**. The amorphization **52** is performed, and the second dopant is introduced to a bottom portion of each of the third openings **193** of the dielectric layer **152** in the third region **13**. The amorphization **52** is as described above in reference to FIG. **13**, and repeated description is omitted herein.

[0054] FIG. **22** is a schematic cross-sectional diagram along the line A-A' shown in FIG. **3** at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. After the amorphization **52**, a portion of the photoresist layer **23** over the second region **12** is removed. An entirety of the second openings **192** in the second region **12** is exposed. In some embodiments, a top view of the photoresist layer **23** at this stage is similar to the top view of the photoresist layer **21** as shown in FIG. **9**.

[0055] FIG. **23** is a schematic cross-sectional diagram along the line A-A' shown in FIG. **3** at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. After exposure of the second openings **192** as shown in FIG. **22**, the operation **603** of the method **600** and/or the operation **704** of the method **700** is performed. The amorphization **51** is performed, and the first dopant is introduced to a bottom portion of each of the third openings **193** in the third region **13** and a bottom portion of each of the second openings **192** in the second region **12** of the dielectric layer **152**. The amorphization **51** is performed as described above, and repeated description is omitted herein.

[0056] FIG. **24** is a schematic cross-sectional diagram along the line A-A' shown in FIG. **3** at a different stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. After the amorphization **51** is performed, the photoresist layer **23** is removed, and all of the openings **19** are exposed. The operation **604** of the method **600** and/or the operation **708** of the method **700** are then performed on the intermediate structure as shown in FIG. **24**. Other operations as shown in FIGS. **15** to **18** are subsequently performed after the operation **604** of the method **600** and/or the operation **708** of the method **700**, to form a structure similar to that shown in FIG. **18**.

[0057] FIGS. **19** to **24** and their related paragraphs above illustrate an alternative embodiment to achieve a purpose of an even etching rate across different regions of the present disclosure. In the embodiments shown in FIGS. **4** to **18** and FIGS. **19** to **24**, the third opening **193** undergoes two amorphizations. In other alternative embodiments, either the third opening **193** or the second opening **192** undergoes only one amorphization, but with different amounts of dopants introduced to the dielectric layer **152**.

[0058] FIGS. **25** to **26** are schematic cross-sectional diagrams along the line A-A' shown in FIG. **3** at different stages of the method **600** and/or the method **700** according to some embodiments of the present disclosure, and FIG. **27** is a top view of a photoresist layer **24** used during the operation shown in FIG. **26**. In some embodiments, the photoresist layer **23** is used during an amorphization **53** (or a third implantation) targeting bottoms of the third openings **193** in the third region as shown in FIG. **25**. In some embodiments, a photoresist layer **24** shown in FIGS. **26** to **27** covering an entirety of the first openings **191** and an entirety of the third openings **193** is used during an amorphization **54** (or a fourth implantation) targeting bottoms of the second openings **192** in the second region **12**. The amorphization **53** and the amorphization **54** may be performed for different

durations and/or at different intensities in order to form a third region **152c** and a second region **152b** with different doping concentrations and/or different structural densities. In some embodiments, a duration of the amorphization **53** is greater than a duration of the amorphization **54**. In some embodiments, an intensity of the amorphization **53** is greater than an intensity of the amorphization **54**. The amorphization **53** can be performed prior to or after the amorphization **54**, and a sequence is not limited herein as long as a same purpose can be achieved. Operations as shown in FIGS. **15** to **18** are subsequently performed on the intermediate structure of FIG. **26** to form a structure similar to that shown in FIG. **18**.

[0059] The method of the present disclosure can be applied in an interconnection structure disposed over electrical components **142** as illustrated above; however, the invention is not limited thereto. The method of the present disclosure can be applied in any stage of a semiconductor manufacturing process when uneven etching rates of trenches, openings or vias across different regions of a substrate are observed. For instance, the method can be applied in formation of contacts **144** as different regions may have different densities of the electrical components **142** and/or different densities of the contacts **144**. FIG. **28** is a schematic cross-sectional diagram along the line A-A' shown in FIG. **3** at a stage of the method **600** and/or the method **700** according to some embodiments of the present disclosure. For case of illustration and simplicity of drawing, only the contacts **144** connected to source/drain structures of the electrical components **142** are depicted. As shown in FIG. **28**, the contacts **144** in different regions **11**, **12** and **13** may have different sizes and/or densities. The method **600** or the method **700** of the present disclosure can be applied in the formation of the contacts **144**, and the contacts **144** in different regions **11**, **12** and **13** may be surrounded by different regions **152a**, **152b** and **152c** with different doping concentrations. In some embodiments, the first region **152a** is dopant-free, and a doping concentration of the second region **152b** is less than a doping concentration of the third region **152c**.

[0060] In accordance with some embodiments of the disclosure, a method for manufacturing a semiconductor structure is provided. The method may include several operations. A first layer is formed over a first region and a second region of a substrate. A first etching is performed on the first layer, thereby forming a first trench in the first region and a second trench in the second region. An amorphization **51** is performed on the first layer in the second region. A second etching is performed on the first layer, wherein an etching rate of the second etching in the second region is greater than an etching rate of the second etching in the first region.

[0061] In accordance with some embodiments of the disclosure, a method for manufacturing a semiconductor structure is provided. The method may include several operations. A dielectric layer is formed over a substrate. Portions of the dielectric layer are removed, thereby forming a first opening in a first region of the substrate, a second opening in a second region of the substrate, and a third opening in a third region of the substrate. A first mask layer is formed in the first region. A first dopant is introduced to a bottom of the second opening and a bottom of the third opening. A second mask layer is formed in the second region. A second dopant is introduced to the bottom of the third opening. The first mask layer and the second mask layer are removed. An etching is performed on the dielectric layer, thereby forming a first through via, a second through via and a third through via.

[0062] In accordance with some embodiments of the disclosure, a semiconductor structure is provided. The semiconductor structure includes a dielectric layer, a first through via, a second through via, and a third through via. The dielectric layer is disposed in an interconnect structure, and includes a first doping region and a second doping region, wherein a first doping concentration of the first doping region is different from a second doping concentration of the second doping region. The first through via is surrounded by a dopant-free region of the dielectric layer. The second through via is surrounded by the first doping region. The third through via is surrounded by the second doping region.

[0063] In accordance with some embodiments of the disclosure, a semiconductor structure is

provided. The semiconductor structure includes a first dielectric layer, a first through via in the first dielectric layer, a second through via in the first dielectric layer, a first doping region disposed in the first dielectric layer, and a second doping region disposed in the first dielectric layer. The second through via is separated from the first through via. The first doping region laterally surrounds a portion of the first through via. The second doping region laterally surrounds a portion of the second through via. The first doping region and the second doping region include a same dopant. A doping concentration of the second doping region is greater than a doping concentration of the first doping region.

[0064] In accordance with some embodiments of the disclosure, a semiconductor structure is provided. The semiconductor structure includes a first dielectric layer, a first through via in the first dielectric layer, a second through via in the first dielectric layer, a first doping region disposed in the first dielectric layer, and a second doping region disposed in the first dielectric layer. The second through via is separated from the first through via. The first doping region laterally surrounds a portion of the first through via. The second doping region laterally surrounds a portion of the second through via. A width of the first through via is less than a width of the second through via. The first doping region includes a first dopant, and the second doping region includes a second dopant different from the first dopant. A concentration of the first dopant is substantially equal to a concentration of the second dopant.

[0065] The foregoing outlines features of several embodiments so that those skilled in the art may better understand aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a dielectric layer, disposed in an interconnect structure, and including a first doping region and a second doping region, wherein a first doping concentration of the first doping region is different from a second doping concentration of the second doping region; a first through via, surrounded by a dopant-free region of the dielectric layer; a second through via, surrounded by the first doping region; and a third through via, surrounded by the second doping region.
2. The semiconductor structure of claim 1, wherein the first through via, the second through via and the third through via are electrically connected to different conductive elements of the interconnect structure.
3. The semiconductor structure of claim 1, wherein the second doping region includes a first dopant same as a dopant of the first doping region, and a second dopant different from the first dopant.
4. The semiconductor structure of claim 1, wherein the first doping region surrounds a bottom portion of the second through via, and the second doping region surrounds a bottom portion of the third through via.
5. A semiconductor structure, comprising: a first dielectric layer over a substrate; a first through via in the first dielectric layer; a first doping region disposed in the first dielectric layer and laterally surrounding a portion of the first through via; a second through via in the first dielectric layer and separated from the first through via; and a second doping region disposed in the first dielectric layer and laterally surrounding a portion of the second through via, wherein the first doping region and the second doping region comprise a same dopant, and a doping concentration of the second doping region is greater than a doping concentration of the first doping region.

6. The semiconductor structure of claim 5, wherein a width of the first through via is less than a width of the second through via.
7. The semiconductor structure of claim 5, wherein the dopant in the first doping region and in the second doping region comprises carbon (C), silicon (Si), germanium (Ge), tin (Sn), lead (Pb), boron (B), aluminum (Al), gallium (Ga), indium (In), thallium (Tl), nitride (N), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi), or a combination thereof.
8. The semiconductor structure of claim 5, further comprising a third through via disposed in the first dielectric layer and separated from the first through via and the second through via.
9. The semiconductor structure of claim 8, wherein a width of the third through via is less than a width of the first through via, and less than a width of the second through via.
10. The semiconductor structure of claim 8, further comprising a dopant-free region disposed in the first dielectric layer and laterally surrounding the third through via.
11. The semiconductor structure of claim 5, wherein the first through via comprises an upper portion and a lower portion coupled to the upper portion, a width of the upper portion is greater than a width of the lower portion.
12. The semiconductor structure of claim 11, wherein the first doping region laterally surrounds the lower portion of the first through via.
13. The semiconductor structure of claim 5, wherein the second through via comprises an upper portion and a lower portion coupled to the upper portion, and a width of the upper portion is greater than a width of the lower portion.
14. The semiconductor structure of claim 13, wherein the second doping region laterally surrounds the lower portion of the second through via.
15. The semiconductor structure of claim 5, further comprises: a second dielectric layer under the first dielectric layer; a first conductive element disposed in the second dielectric layer; and a second conductive element disposed in the second dielectric layer and separated from the first conductive element, wherein a width of the first conductive element is less than a width of the second conductive element.
16. The semiconductor structure of claim 15, wherein the first through via is coupled to the first conductive element, and the second through via is coupled to the second conductive element.
17. The semiconductor structure of claim 5, further comprises: a plurality of first electrical elements disposed over the substrate; and a plurality of second electrical elements disposed over the substrate and separated from the plurality of first electrical elements, wherein a density of the first electrical elements is greater than a density of the second electrical elements.
18. The semiconductor structure of claim 17, wherein the first through via is coupled to one of the plurality of first electrical elements, and the second through via is coupled to one of the plurality of second electrical elements.
19. A semiconductor structure, comprising: a first dielectric layer; a first through via in the first dielectric layer; a first doping region disposed in the first dielectric layer and laterally surrounding a portion of the first through via; a second through via in the first dielectric layer and separated from the first through via; and a second doping region disposed in the first dielectric layer and laterally surrounding a portion of the second through via, wherein a width of the first through via is less than a width of the second through via, wherein the first doping region comprises a first dopant, and the second doping region comprises a second dopant different from the first dopant, and a concentration of the first dopant is substantially equal to a concentration of the second dopant.
20. The semiconductor structure of claim 19, further comprises: a second dielectric layer under the first dielectric layer; a first conductive element disposed in the second dielectric layer; and a second conductive element disposed in the second dielectric layer and separated from the first conductive element, wherein a width of the first conductive element is less than a width of the second

conductive element. wherein the first through via is coupled to the first conductive element, and the second through via is coupled to the second conductive element.
